BE Smart Cities - Dynamic Models for Building Energy Management

KAHAN Joachim
LECOMPTE-BOINET Simon
NGUYEN ANH Huy
TRAN ANH Phi

May 14, 2026

1 Modeling

1.1 Geometrical description

This project involves a thermal analysis of a room in a city flat. The room under study is rectangular, measuring 4 metres by 5 metres by 3 metres ("Room 1" in Figure 1), which corresponds to a total area of 20 m^2 and a volume of 60 m^3. This model takes into account the four walls of the room, each with distinct boundary conditions:

- **Wall 1 (Exterior, 5m x 3m)**: Connected to the outside (T_0), composed of brick (0.2 m) and insulation (0.2 m), and including a 2 m^2 window.
- **Wall 2 (Exterior, 4m x 3m)**: Connected to the outside (T_0), composed of brick (0.2 m) and insulation (0.2 m), without any opening.
- **Wall 3 (Interior, 5m x 3m)**: Connected to a corridor (T_2), composed solely of brick (0.2 m), and including a 2 m^2 door.
- **Wall 4 (Interior, 4m x 3m)**: Connected to a neighboring heated room (T_1), composed solely of brick (0.2 m).

Furthermore, behind walls 1 and 2 is the exterior of the building, which will be at temperature T_0 (which varies), wall 3 separates the room from the building's corridor, which will have a temperature of T_2 (17°C – 19°C), and finally wall 4 separates the room from another bedroom, which will have a temperature of T_1 (17°C – 19°C).

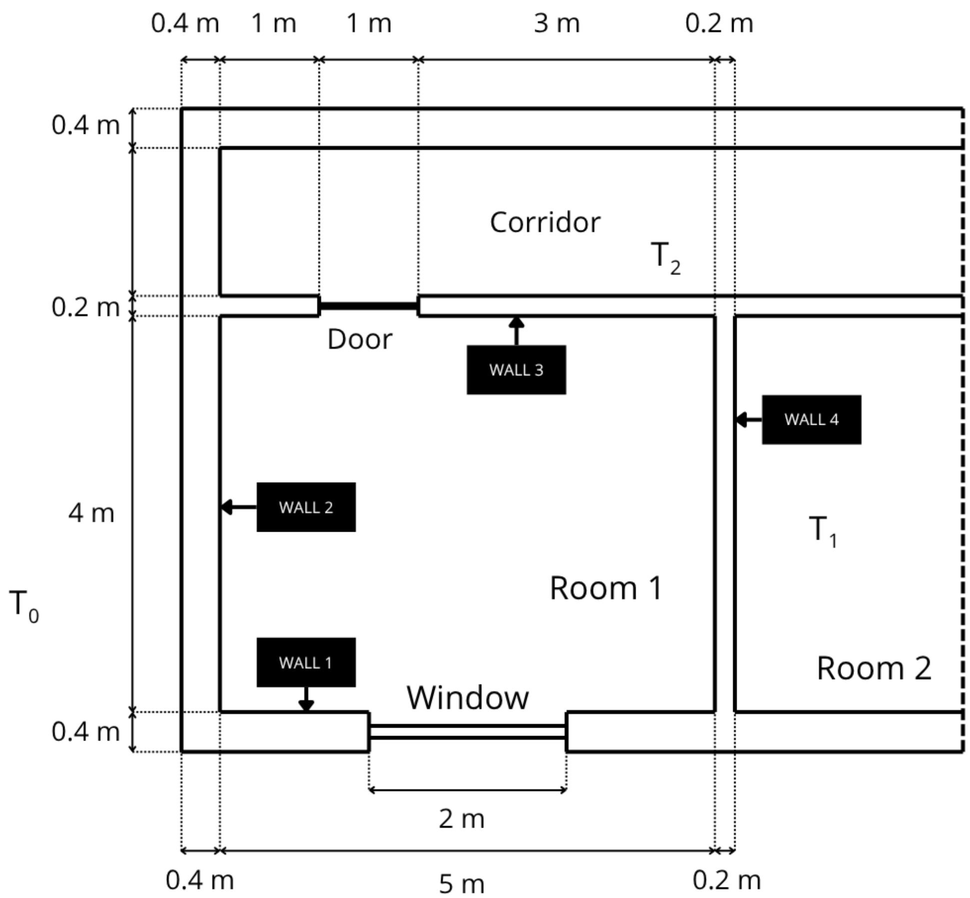

Figure 1: Geometrical diagram of the room with the 4 walls

1.2 Hypothesis

- The temperature is uniform on each surface of a wall.
- Heat transfers with the floor and ceiling are neglected.
- Heat transfers are assumed to be linear.
- Long wave radiation between the inner surfaces is neglected.
- Convection coefficients are as follows: Indoor air $h = 8\,W/(m^2K)$, Outdoor air $h = 25\,W/(m^2K)$.
- Air exchange rate: 0.5 air changes/hour

2 Material Properties

2.1 Composition of the walls 1 & 2

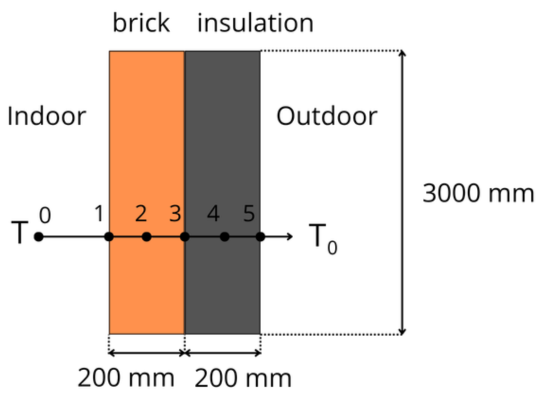

Figure 2: Walls 1 & 2 schematic

The walls 1 and 2 have the same composition, as the figure 2 describe it, they are composed with 200 mm width of brick and 200 mm width of insulation. This wall separates the outside from the inside and it is 3 m high.

2.2 Composition of the window

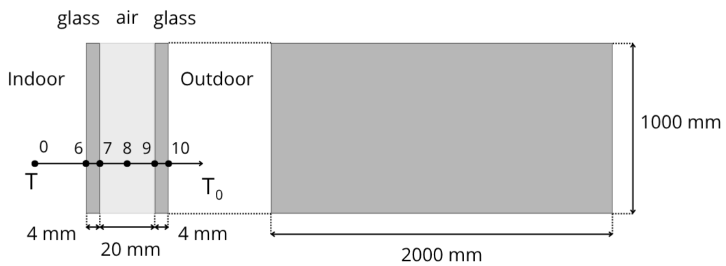

Figure 3: Window schematic

The window is in the middle of the wall 1, it is composed of double glazing : both glass are 4 mm width and between there is 20 mm gap fill with air. The window is 2 m wide and 1 m high.

2.3 Composition of the walls 3 & 4

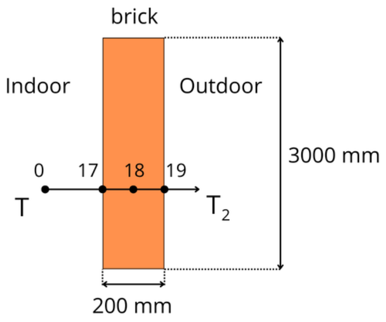

Figure 4: Wall 3 schematic

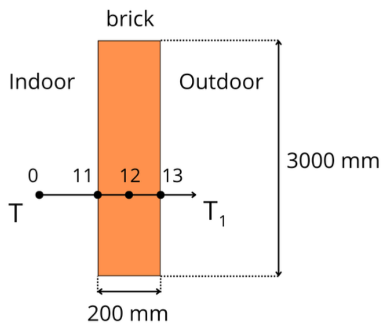

Figure 5: Wall 4 schematic

The walls 3 and 4 are the same : they are composed with 200 mm width of brick and they are 3 m high.

2.4 Composition of the door

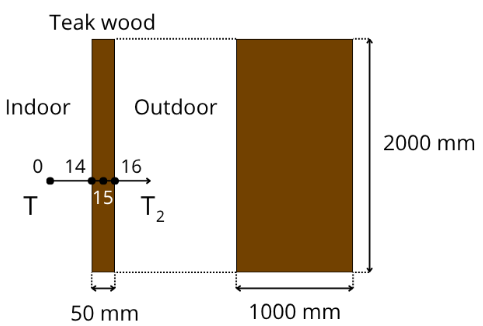

Figure 6: Door schematic

And finally, The door separates the corridor from the room, it is 50 mm width, 1 m wide and 2 m high.

2.5 Reminder

The following table summarizes the thermo-physical characteristics of the materials used in our model:

Table 1: Characteristics of materials[1]

Material	**Conductivity** $(W/m \cdot K)$	**Specific Heat** $(J/kg \cdot K)$	**Density** (kg/m^3)	**Width** (mm)
Brick wall	0.9	900	1800	200
Fibre board	0.048	2100	16	200
Air	0.026	1000	1.2	N/A
Glass (avg prop)	1.046	837	2300	4
Teak wood	0.172	2301	640	50

3 Wall and Window Property

Table 2: Characteristics of wall[2]

Albedo	$\boldsymbol{\alpha_o}$	$\boldsymbol{\alpha_i}$	$\boldsymbol{\beta}$	$\boldsymbol{\gamma}$
0.25	0.3	0.35	90	50.8

Albedo: Albedo coefficient

α_o**:** Absorptance of the outdoor surface of the wall in short wave

α_i**:** Absorptance of the indoor surface of the wall in short wave

β**:** Tilt angle compared to the horizontal (°)

γ**:** Azimuth angle (°)

S**:** Wall surface area (m^2)

For the calculation of irradiance, the two sides of wall 1 are merged into one, where a and b represent the length of each side. The resulting length l is calculated as:

$$l = \sqrt{a^2 + b^2} = 7 \tag{1}$$

Table 3: Characteristics of window[3]

$\boldsymbol{\alpha_w}$	τ_w
0.3	0.35

α_w**:** Absorptance of the outdoor surface of the window in short wave

τ_w**:** Transmittance of the window in short wave

The window is assumed to be a high-performance tinted one, hence the low value of absorptance and transmittance.

4 Thermal Network

4.1 System description

The new thermal model is based on the finite volume method (FVM). To accurately represent the dynamics of solid and composite walls, the network has been extended to **20 nodes and 26 branches** according to the provided circuit diagram. The interior components have been modeled precisely following the RC network layout.

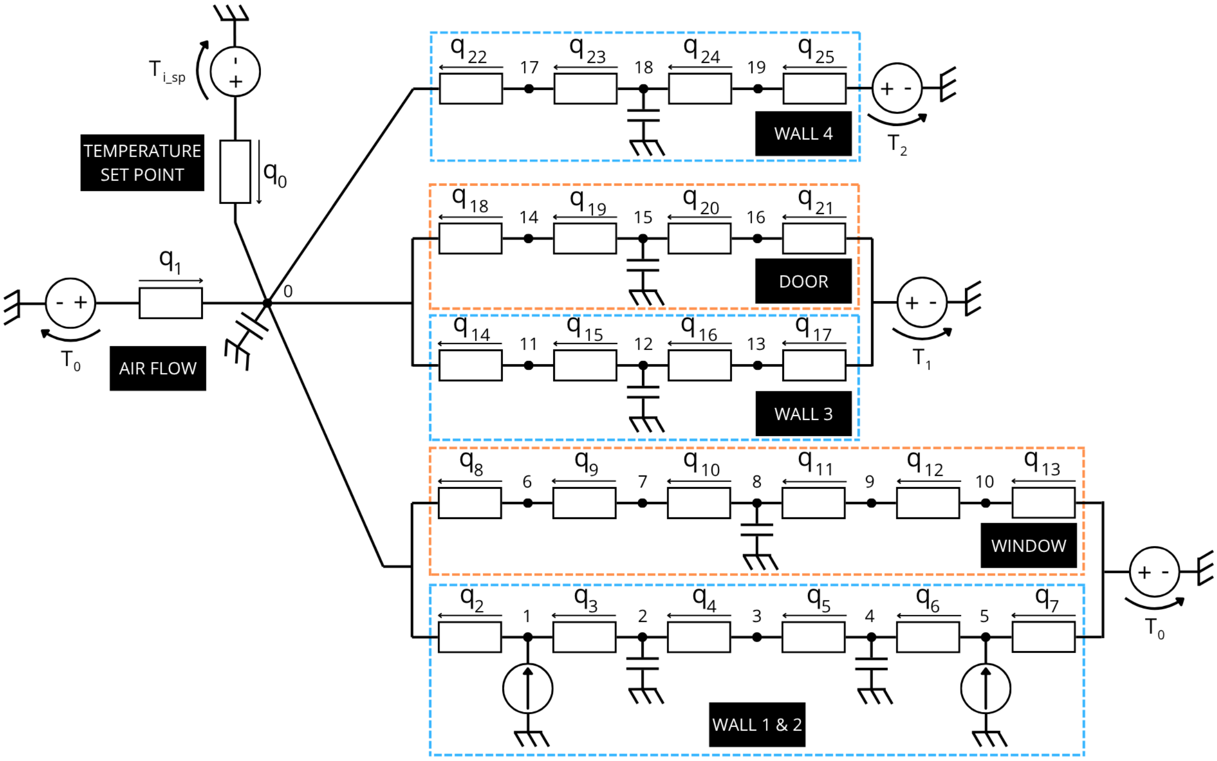

Figure 7: Diagram of the thermal network (20 nodes, 26 branches)

4.2 System matrices

Due to their large dimensions required to accurately model the complex system, the incidence (A, size 26x20), conductance (G, size 26x26), and capacity (C, size 20x20) matrices calculated by our model are extracted and presented below:

```
INCIDENCE MATRIX A (26 branches x 20 nodes)
============================================================
[[ 1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0. 0.  0.  0.  0.  0.]
 [ 1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
 [ 1. -1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
 [ 0.  1. -1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
 [ 0.  0.  1. -1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
 [ 0.  0.  0.  1. -1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
 [ 0.  0.  0.  0.  1. -1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
 [ 0.  0.  0.  0.  0.  1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
 [ 1.  0.  0.  0.  0.  0. -1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
 [ 0.  0.  0.  0.  0.  0.  1. -1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.  1. -1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.  0.  1. -1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.  0.  0.  1. -1.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  1.  0.  0.  0.  0.  0.  0.  0.  0.  0.]
 [ 1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0. -1.  0.  0.  0.  0.  0.  0.  0.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  1. -1.  0.  0.  0.  0.  0.  0.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  1. -1.  0.  0.  0.  0.  0.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  1.  0.  0.  0.  0.  0.  0.]
 [ 1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0. -1.  0.  0.  0.  0.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  1. -1.  0.  0.  0.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  1. -1.  0.  0.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  1.  0.  0.  0.]
 [ 1.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0. -1.  0.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  1. -1.  0.]
 [ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  1. -1.]
 [ 0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  0.  1.]]
```

Figure 8: Incidence matrix A values

Values of matrices G and C

```
=================================================================
CONDUCTANCE MATRIX G (26 x 26) - Diagonal (Non-zero elements)
=================================================================
G[0,0]=1000.0   G[1,1]=20.0     G[2,2]=200.0    G[3,3]=112.5
   G[4,4]=112.5
G[5,5]=12.0     G[6,6]=12.0     G[7,7]=625.0    G[8,8]=16.0
   G[9,9]=2800.0
G[10,10]=2800.0 G[11,11]=2800.0 G[12,12]=2800.0 G[13,13]=50.0
   G[14,14]=104.0
G[15,15]=58.5   G[16,16]=58.5   G[17,17]=104.0  G[18,18]=16.0
   G[19,19]=16.0
G[20,20]=16.0   G[21,21]=16.0   G[22,22]=96.0   G[23,23]=54.0
   G[24,24]=54.0   G[25,25]=96.0


=================================================================
CAPACITY MATRIX C (20 x 20) - Diagonal (Non-zero elements)
=================================================================
C[0,0]=72000.0       C[2,2]=16200000.0    C[4,4]=168000.0
C[8,8]=15000.0       C[12,12]=8424000.0   C[15,15]=64000.0
C[18,18]=7776000.0
```

(Note: For better readability, only the non-zero diagonal elements of matrices G and C have been listed explicitly.)

Python Code for Vectors and Steady-State Calculation

```
# ==========================
# 5. VECTORS b AND f
# ==========================
b = np.zeros(nq)
b[0] = Trad      # q0 source (HVAC)
b[1] = To        # q1 source (Ventilation)
b[7] = To        # End of Wall 1
b[13] = To       # End of Window
b[17] = T1       # End of Wall 2
b[21] = T1       # End of Door
b[25] = T2       # End of Wall 3

f = np.zeros(n  )
f[1] = tau_g * alpha_i * Sg * E     # Node 1 (Solar internal)
f[5] = alpha_o * Sw1 * E            # Node 5 (Solar external)

# ==========================
# 6. STEADY-STATE CALCULATION
# ==========================
# System matrix K = A^T * G * A
K = A.T @ G @ A

# Global load vector (forcing term)
load_vector = A.T @ G @ b + f

# Solving the linear system K * theta = load_vector
theta = np.linalg.solve(K, load_vector)
```

5 Steady-state test in worst case scenarios

The section aims to examine the model's behavior in extreme weather conditions: very high or very low temperature. The steady-state calculation follows the methodology mentioned in the previous section.

5.1 Hypothesis

The controller is simulated to be near perfect, with a conductance value of:

$$G = 1000 \text{ W/K}$$

The worst-case scenarios are defined as the highest and lowest temperatures recorded in Paris Orly since 1955. The extreme high temperature scenario is assumed to occur during the day, and the extreme low temperature scenario is assumed to occur during the night. Consequently, the temperature setpoint and the surrounding indoor areas for the former case are higher than the latter. The parameters for both scenarios are detailed below:

Table 4: Parameters for Extreme High and Extreme Low Temperature Scenarios

Parameter	**Symbol**	**Extreme High**	**Extreme Low**
Outdoor temperature	T_o	41.9 °C	-16.8 °C
Setpoint temperature	$T_{i,sp}$	19 °C	17 °C
Temperature of corridor	T_1	19 °C	17 °C
Temperature of next room	T_2	19 °C	17 °C
Solar irradiance	E	1000 W/m^2	0 W/m^2

5.2 Result

Table 5: HVAC Performance under Extreme Temperature Scenarios

Parameter	**Extreme High**	**Extreme Low**
Indoor air temperature (°C)	20.13	15.83
Difference from setpoint (°C)	1.13	1.17
HVAC power (W)	1130	1170

The result shows that in both scenarios, the deviation from the temperature setpoint is larger than 1 °C. The extreme low temperature scenario is slightly more detrimental to HVAC operation, as calculations yield both a higher difference from the temperature setpoint and higher HVAC power consumption.

6 Yearly thermal load

This test simulates the building's behavior subjected to variable weather conditions (outdoor temperature and solar irradiance) while maintaining a variable indoor temperature setpoint (day/night).
In order to obtain the yearly thermal load, the building is simulated for the first 6 days of January, March, May, July, September and November. The total thermal load of the period is then multiplied to represent the whole month and the whole year.
The controller is simulated to be near perfect, with a conductance value of:

$$G = 1000 \text{ W/K}$$

6.1 Weather input data

The weather data for a year is obtained via the .epw file available on Energyplus website for Paris Orly airport[4]. The temperature setpoint and the temperature of the surrounding area is defined as 19 °C for the day (6h00-22h00) and 17 °C for the night.

Input of weather data

```
filename = '../weather_data/FRA_Paris.Orly.071490_IWEC.epw'
[data, meta] = dm4bem.read_epw(filename, coerce_year=None)
weather = data[["temp_air", "dir_n_rad", "dif_h_rad"]]
```

6.2 Time step

The eigenvalue analysis of the state matrix A_s allowed us to determine the maximum allowable time step for explicit Euler integration, as well as the settling time of the system.

Eigenvalue analysis for obtaining time step

```
[As, Bs, Cs, Ds, us] = dm4bem.tc2ss(TC) # transform system to state
    space
alpha= np.linalg.eig(As)[0]  # Eigenvalue of As
deltatmax = 2 * min(-1. / alpha)    # max time step for stability of
    Euler explicit
dt = dm4bem.round_time(deltatmax) #Round time down to minutes
deltatmax = 121 s = 2.0 min
dt = 120 s = 2.0 min
```

For this simulation, dt:1.0 min is used instead, to improve the numerical stability of the model.

6.3 Time integration

The simulation uses explicit Euler method for time integration.

Explicit Euler method

```
I = np.eye(As.shape[0])
for k in range(u.shape[0] - 1):
theta.iloc[k + 1] = (I + dt * As) @ theta.iloc[k] + dt * Bs @ u.iloc[k]
```

6.4 Result

The following graph shows the result for January and July, in which θ_{indoor} indicates the indoor air temperature, θ_{outdoor} indicates the outdoor air temperature and q_{HVAC} indicates the HVAC thermal load.

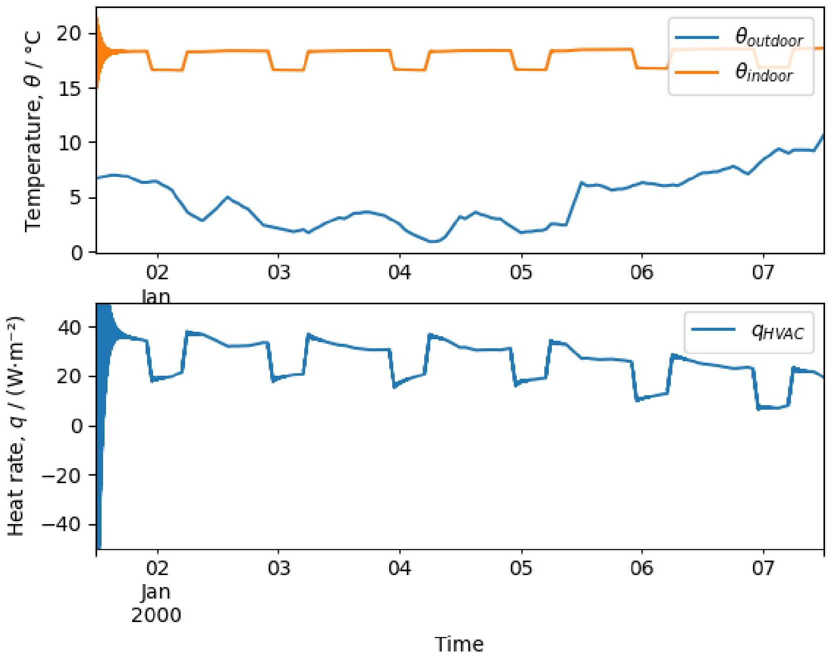

Figure 9: Indoor, outdoor temperature and thermal load in January

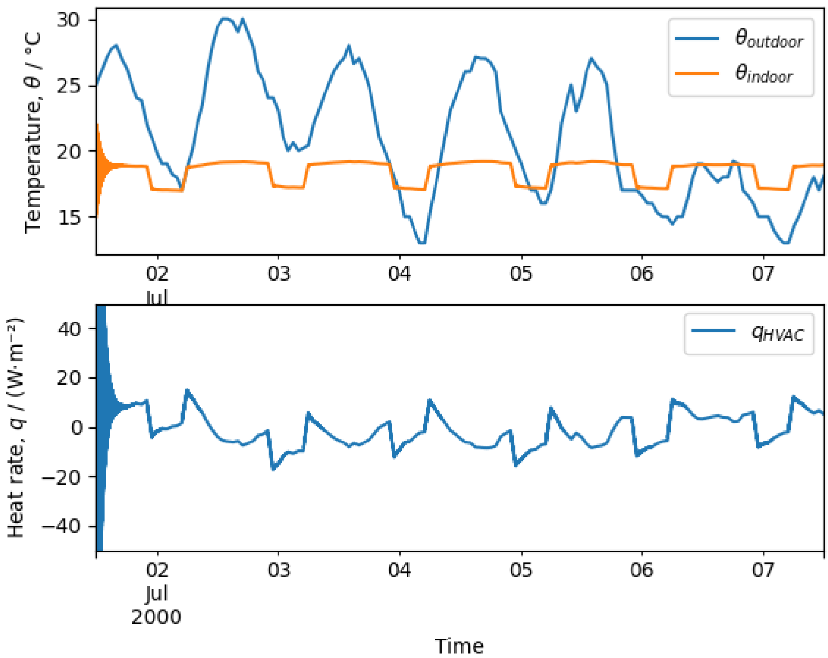

Figure 10: Indoor, outdoor temperature and thermal load in July

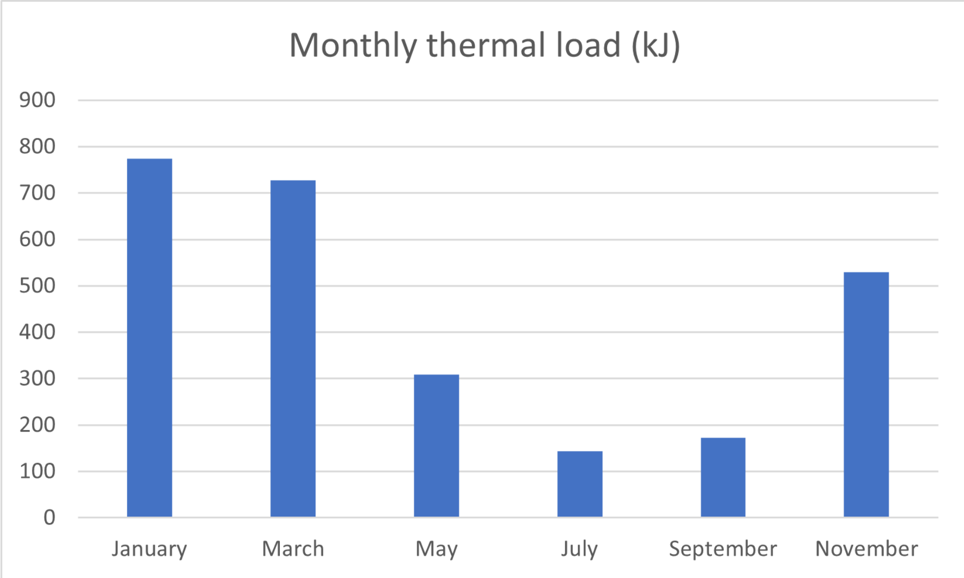

Figure 11: Thermal load by month

Table 6: 5-day, monthly and yearly thermal load

Month	**5 days (J/m^2)**	**Full month (MJ)**
January	6244823.24	774.3580817
March	5864444.306	727.1910939
May	2485311.257	308.1785958
July	1162230.612	144.1165959
September	1391602.366	172.5586934
November	4268051.851	529.2384296
Total		5311.282981
Total (kWh)		1475.356384

It can be seen from the time-series graphs that the model experiences significant instability at the beginning but achieves stability afterward. Thus, for monthly thermal load calculation, the result of the first day is omitted, and only the last 5 days are taken into account. The result shows that the climate of May to September allows for minimal HVAC operation, whereas HVAC usage intensifies between November and March. It is also worth noting that in the cold months such as January and March, all of HVAC operation is for heating, whereas in July, most of the operation is for cooling. Meanwhile, in May and September, most of HVAC usage is for cooling, but at a lower intensity than between November and January.

7 Nonlinear Controller Analysis

7.1 Free Running in Winter

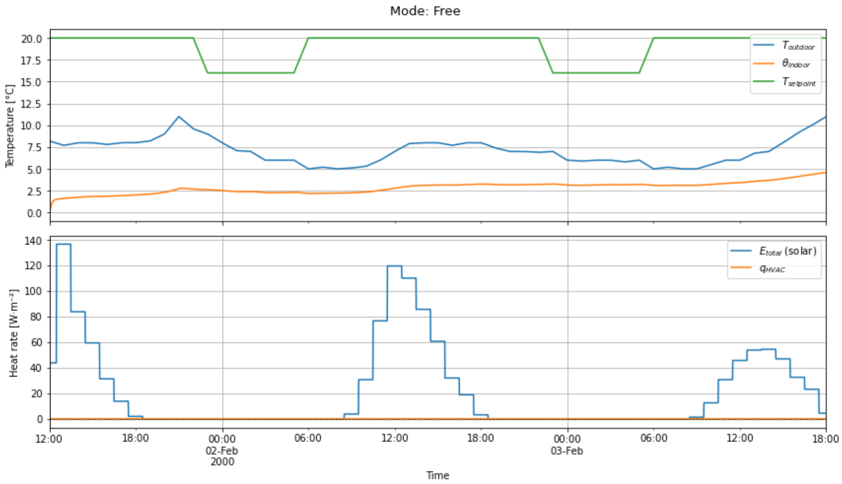

Figure 12: Free running in February

In this mode, the HVAC system is completely inactive, and the q_{HVAC} line remains constant at 0 throughout the period. Indoor temperature θ_i falls well below the day-time setpoint (20 °C) during the night, hovering between 0 and 5 degrees, illustrating the need for heating in winter.

Solar radiation (E_{tot}) is low (winter, mostly diffuse radiation), so solar gains have little effect on raising the indoor temperature. Without control, the building is unable to maintain comfort during cold periods.

7.2 Heating Only in Winter

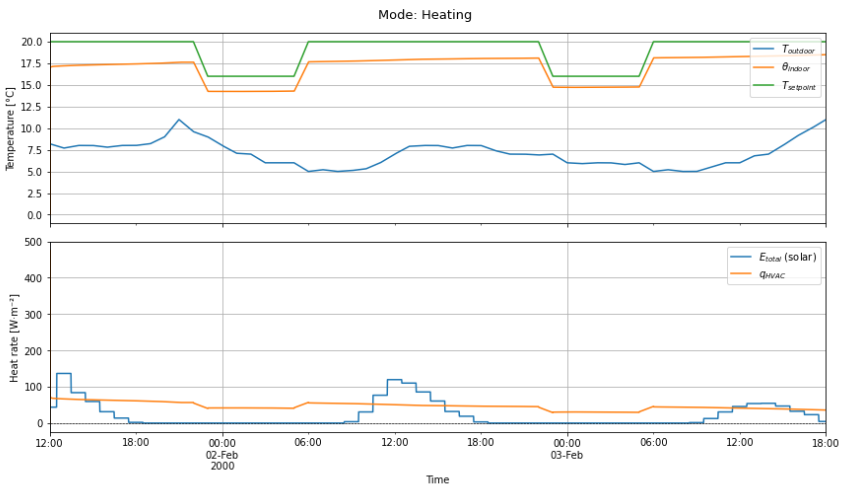

Figure 13: Heating in February

Indoor temperature θ_i is successfully kept close to the setpoint $T_{i,sp}$ (16 °C at night, 20 °C during the day). The heating power q_{HVAC} is always positive, acting only when $\theta_i < T_{i,sp}$ (proportional control).
Some overshoot is visible after the morning setpoint rise, and the controller continues heating until the error is reduced. Proportional heating can maintain thermal comfort, but may cause temporary overshoot if the gain is high.

7.3 Cooling Only in Summer

Based on the Paris data, outdoor temperatures in July exceed 25 °C, and solar radiation is strong. Although it appears that cooling is not needed during this time, given that q_{HVAC} is zero or negligible throughout this period.

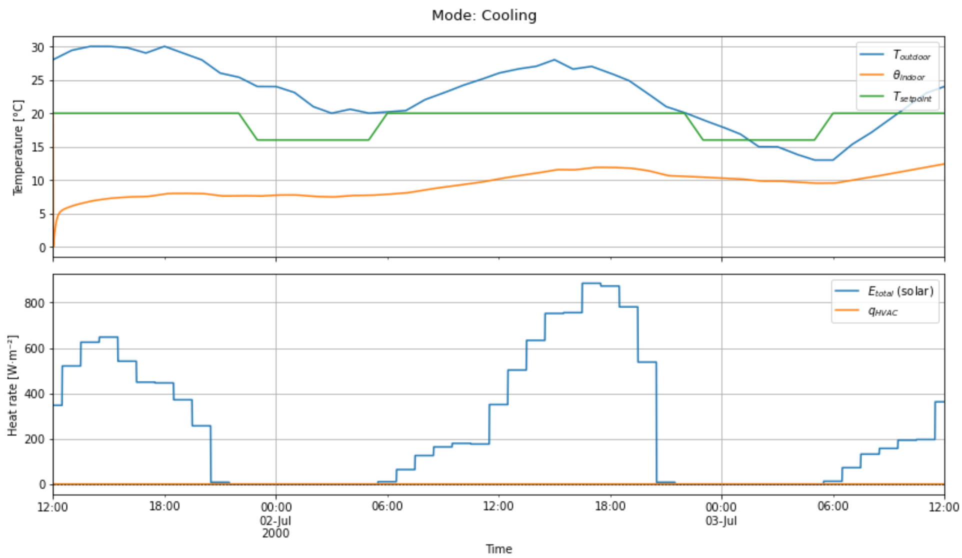

Figure 14: Cooling in July with higher temperature setpoints

Cooling is only activated when θ_i exceeds $T_{i,sp}$ + DEADBAND (20 + 4 = 24 °C). The deadband prevents frequent on-off cycling. In this case, because the indoor temperature stays well below the 20 degrees setpoint, the cooling does not need to be activated.
If we change the setpoint to 10 degrees during the day and 6 degrees at night, we can observe that q_{HVAC} is negative (cooling is activated) and roughly proportional to the error. Cooling-only control with a deadband avoids unnecessary chiller operation while limiting indoor overheating.

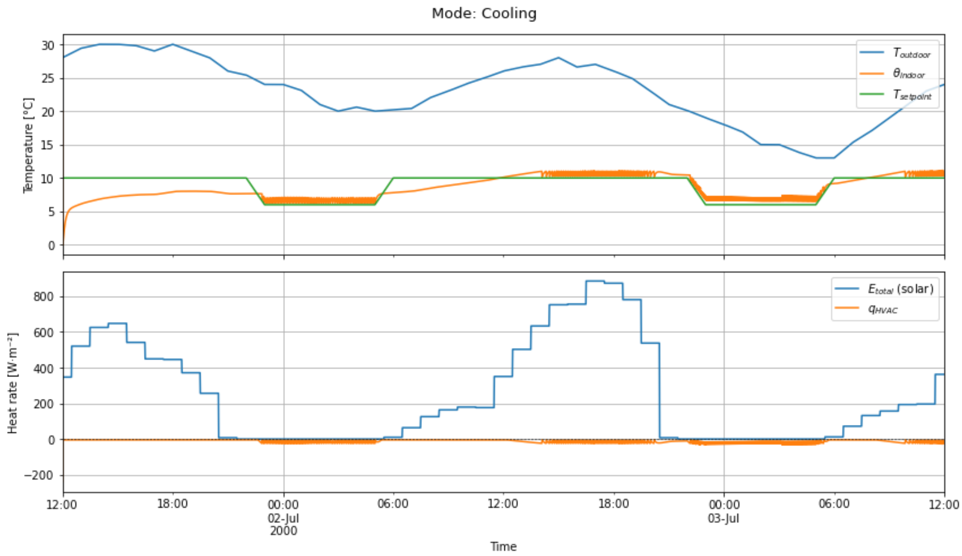

Figure 15: Cooling in July with lower temperature setpoints

7.4 Heating + Cooling with Deadband in Spring

Both heating and cooling are available, but the controller is inactive when θ_i lies between $T_{i,sp}$ and $T_{i,sp}$ + DEADBAND, allowing the temperature to float within the deadband range.

There is a small pulse of heating from q_{HVAC} early in the morning to prevent temperature from dropping too far below the setpoint. The solar gain is high, around $900\,\mathrm{W/m^2}$, allowing indoor temperature to remain comfortable around 18 °C. Combined heating/-cooling with a deadband provides comfortable temperatures year-round while limiting unnecessary HVAC activity.

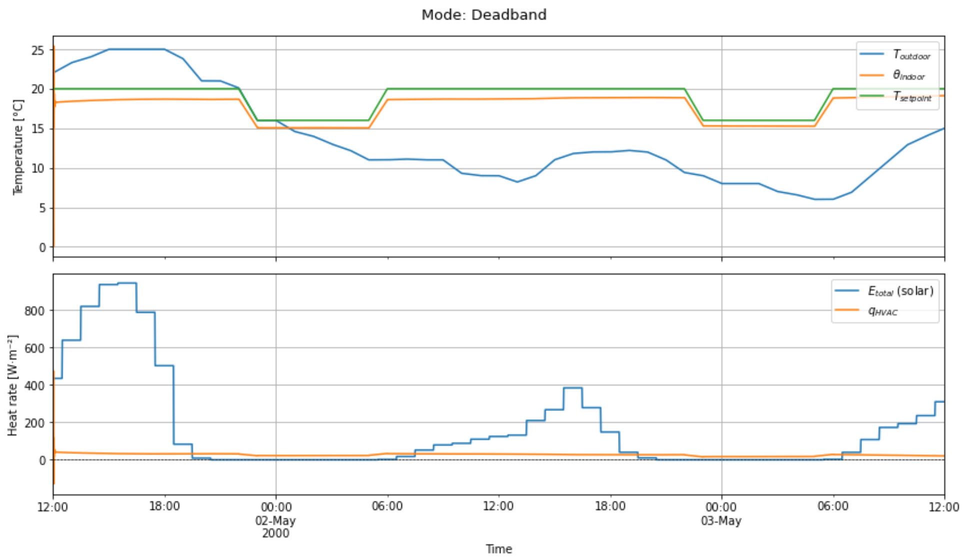

Figure 16: Both heating and cooling with deadband

7.5 Solar Protection in Spring

When θ_i exceeds $T_{i,sp} + 1\,°\mathrm{C}$, the shutters close. The solar radiation peaks are reduced from roughly 900 to 400 $\mathrm{W/m^2}$ to simulate the use of blinds or curtains.
Because solar gain is restricted, the indoor temperature rises slowly and remains lower when compared to deadband mode. Cooling is not necessary ($q_{HVAC} = 0$); the shutters do most of the work by cutting the solar load. Active solar protection (shutters) is an effective passive cooling strategy that reduces the mechanical cooling demand.

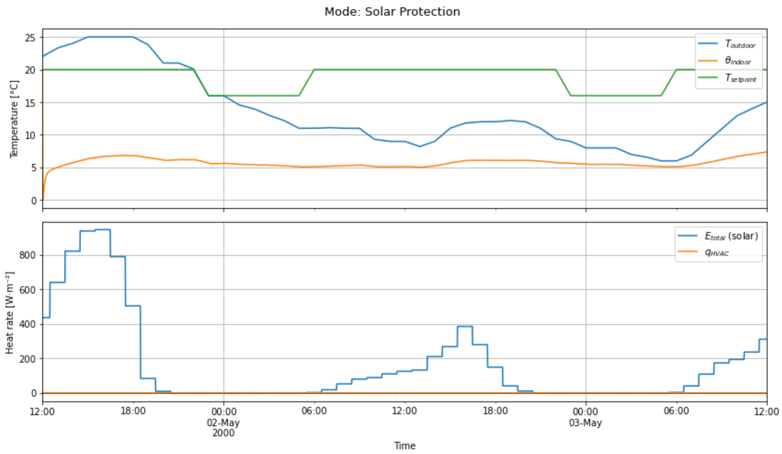

Figure 17: Solar protection

7.6 Passive Cooling in Spring

Passive cooling typically involves using natural elements like night ventilation rather than mechanical HVAC. Here, the shutters close when $\theta_i > T_{i,sp} + 1\,°\mathrm{C}$ (same as solar protection). Additionally, when the outdoor air is cooler than the setpoint, free-cooling ventilation (ACH = 1) is enabled. Despite the high solar gain, the indoor temperature remains low, likely due to high thermal mass or night air flushing keeping the heat from accumulating.

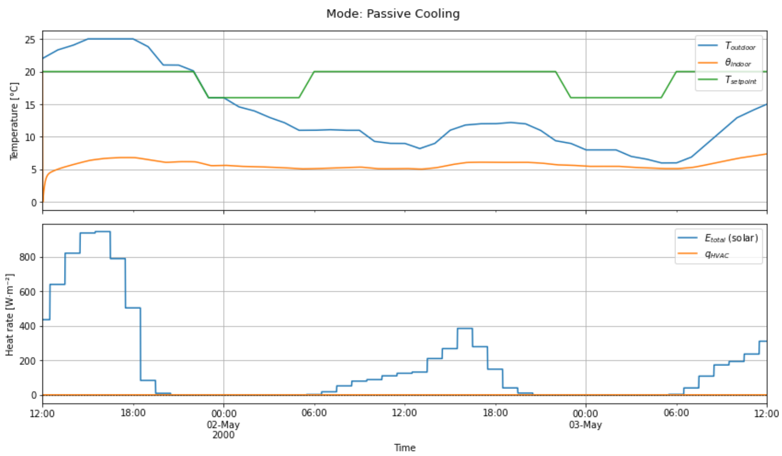

Figure 18: Passive cooling

8 Conclusion

The thermal network model successfully captures the dynamic behavior of the room, and the steady-state and yearly simulations confirm that the HVAC system can maintain acceptable indoor comfort under extreme weather conditions, with deviations of about 1.1–1.2°C and an annual thermal load of approximately 1475kWh. Controller analyses further show that while free running leads to discomfort in winter, heating-only and cooling-only strategies can maintain setpoints when properly tuned. Adding a deadband reduces unnecessary HVAC cycling, and passive measures—such as solar protection (shutters) and night ventilation—significantly limit overheating without mechanical cooling. Overall, combining a deadband controller with passive cooling strategies effectively maintains thermal comfort year-round while minimising HVAC energy use.

References

[1] Thermtest. "Thermal properties of materials database. "[Online]. Available: https://thermtest.com/thermal-resources/materials-database. (Accessed: 2026-05-14).

[2] The Engineering ToolBox. "Solar radiation absorptivity. "[Online]. Available: https://www.engineeringtoolbox.com/radiation-surface-absorptivity-d_1805.html. (Accessed: 2026-05-14).

[3] Ingener. "Cabin thermal loads: Solar radiation loads. "[Online]. Available: https://ingener.by/specialty-applications-testing/specialty-hvac-applications/automobiles-hvac/cabin-thermal-loads/solar-radiation-loads. (Accessed: 2026-05-14).

[4] EnergyPlus. "Weather data. "[Online]. Available: https://energyplus.net/weather. (Accessed: 2026-05-14).